

Junseok Lee

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RESEARCH INTEREST

I am interested in researching general machine-learning methodologies and applying them to real-world problems, with a particular focus on single-cell RNA analysis and protein design.

- Graph Representation Learning
- Single-cell RNA-seq analysis
- Protein Design

EDUCATION

KAIST (Korea Advanced Institute of Science and Technology)

- Ph.D. in Industrial and Systems Engineering Mar 2023 – Present
 - Research Interest: Graph Representation Learning, Bioinformatics
 - Advisor: [Prof. Chanyoung Park](#)

KAIST (Korea Advanced Institute of Science and Technology)

- M.S. in Industrial and Systems Engineering Mar 2021 – Feb 2023
 - Research Interest: Graph Representation Learning
 - Advisor: [Prof. Chanyoung Park](#)

Pusan National University

- B.S. in Industrial Engineering Mar 2015 – Feb 2021
 - GPA: 4.15/4.5

POSITIONS

University of Texas Health Science Center at Houston (UTHealth)

- Research Assistant at McWillaims School of Biomedical Informatics Jul 2024 – Feb 2025
 - Advisor: [Prof. Yejin Kim](#)

PUBLICATIONS

(†: Equal contribution)

CONFERENCES

- [C9] Oldie but Goodie: Re-illuminating Label Propagation on Graphs with Partially Observed Features
Sukwon Yun, Xin Liu, Yunhak Oh, **Junseok Lee**, Tianlong Chen, Tsuyoshi Murata, Chanyoung Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025**)
- [C8] Global Context-aware Representation Learning for Spatially Resolved Transcriptomics
Yunhak Oh[†], **Junseok Lee**[†], Yeongmin Kim, Sangwoo Seo, Namkyeong Lee, Chanyoung Park
International Conference on Machine Learning (**ICML 2025**)
- [C7] Subgraph Federated Learning for Local Generalization
Sungwon Kim, Yoonho Lee, Yunhak Oh, Namkyeong Lee, Sukwon Yun, **Junseok Lee**, Sein Kim, Carl Yang, Chanyoung Park
International Conference on Learning Representations (**ICLR 2025 Oral Presentation**)
- [C6] Task-Equivariant Graph Few-shot Learning
Sungwon Kim, **Junseok Lee**, Namkyeong Lee, Wonjoong Kim, Seungyeon Choi, Chanyoung Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2023**)
- [C5] Conditional Graph Information Bottleneck for Molecular Relational Learning
Namkyeong Lee, Dongmin Hyun, Gyoung S. Na, Sungwon Kim, **Junseok Lee**, Chanyoung Park
International Conference on Machine Learning (**ICML 2023**)
- [C4] Heterogeneous Graph Learning for Multi-modal Medical Data Analysis
Sein Kim, Namkyeong Lee, **Junseok Lee**, Dongmin Hyun, Chanyoung Park
AAAI Conference on Artificial Intelligence (**AAAI 2023 Oral Presentation**)
- [C3] Relational Self-Supervised Learning on Graphs
Namkyeong Lee, Dongmin Hyun, **Junseok Lee**, Chanyoung Park
ACM International Conference on Information and Knowledge Management (**CIKM 2022**)

- [C2] GraFN: Semi-Supervised Node Classification on Graph with Few Labels via Non-Parametric Distribution Assignment
Junseok Lee, Yunhak Oh, Yeonjun In, Namkyeong Lee, Dongmin Hyun, Chanyoung Park
 ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2022 Short Paper**)
- [C1] Augmentation-Free Self-Supervised Learning on Graphs
 Namkyeong Lee, **Junseok Lee**, Chanyoung Park
 AAAI Conference on Artificial Intelligence (**AAAI 2022**)

JOURNALS

- [J3] Single-cell RNA Sequencing Data Imputation Using Bi-level Feature Propagation
Junseok Lee[†], Sukwon Yun[†], Yeongmin Kim, Tianlong Chen, Manolis Kellis, Chanyoung Park
Briefings in Bioinformatics (2024)
- [J2] Deep Single-cell RNA-seq data Clustering with Graph Prototypical Contrastive Learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, Chanyoung Park
Bioinformatics (2023)
- [J1] Self-Supervised Graph Representation Learning via Positive Mining
 Namkyeong Lee, **Junseok Lee**, Chanyoung Park
Information Sciences (2022)

WORKSHOPS

- [W4] Capturing functional context of genetic pathways through hyperedge disentanglement
 Yoonho Lee, **Junseok Lee**, Sangwoo Seo, Sungwon Kim, Yeongmin Kim, Chanyoung Park
 ICLR Workshop on Machine Learning for Genomics Explorations (**ICLR MLGenX 2025**)
- [W3] Subgraph Federated Learning for Local Generalization
 Sungwon Kim, Yoonho Lee, Carl Yang, Yunhak Oh, Namkyeong Lee, Sukwon Yun, **Junseok Lee**, Sein Kim, Chanyoung Park
 KDD Workshop on Federated Learning for Data Mining and Graph Analytics (**FedKDD 2024**) (**Best Paper, Oral**)
- [W2] Single-cell RNA-seq data imputation using Feature Propagation
 Sukwon Yun[†], **Junseok Lee**[†], Chanyoung Park
 ICML Workshop on Computational Biology (**WCB 2023**) (**Best Paper, Contributed Talk**)
- [W1] Deep Single-cell RNA-seq data Clustering with Graph Prototypical Contrastive Learning
Junseok Lee, Sungwon Kim, Dongmin Hyun, Namkyeong Lee, Yejin Kim, Chanyoung Park
 ICML Workshop on Computational Biology (**WCB 2023**)

PROJECTS	Development of Artificial Intelligence-Based Insurance Claims Payout Prediction Model ■ Collaboration with Shinhan Life	2021
AWARDS	■ KAIST Graduate Student Outstanding Paper Award ■ Best Paper Award at KDD 2024 Workshop on Federated Learning for Data Mining and Graph Analytics ■ Best Paper Award at ICML 2023 Workshop on Computational Biology	Oct 2024 Aug 2024 Jul 2023
TEACHING EXPERIENCE	Teaching Assistant ■ DS503: Special Topics in Data Science <Data Engineering for Big Data Analytics> ■ IE343: Statistical Machine Learning ■ KSE801: Special Topics in Knowledge Service Engineering <Recommender System and Graph Machine Learning>	Spring 2024 Spring 2022 Fall 2022
PROFESSIONAL SERVICES	Area Chair ■ NeurIPS Workshop (AI for Science: from Theory to Practice) [2023]	

- ICML Workshop (AI for Science: Scaling in AI for Scientific Discovery) [2024]

Conference Reviews

- Conference on Neural Information Processing Systems (NeurIPS) [2025]
- The International Conference on Learning Representations (ICLR) [2025]
- The International Conference on Machine Learning (ICML) [2025]

Journal Reviews

- Briefings in Bioinformatics (BiB)
- npj Artificial Intelligence
- BMC Genomics
- Evolving Systems (EVOS)
- International Journal of Machine Learning and Cybernetics (IJMLC)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Cognitive and Developmental Systems (TCDS)
- Artificial Intelligence Review

TALKS AND SEMINARS

Single-cell RNA-seq data imputation using Feature Propagation

- Contributed talk on ICML Workshop on Computation Biology (WCB) 2023

REFERENCES

Prof. Chanyoung Park

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